

TEACHING & RESEARCH DESIGN

SCENARIO 2 - GLOBAL TEAM, SHIFTING REQUIREMENTS

THE STORY

LearnLoop is a startup that just raised 3.2M EUR to build an AI-powered adaptive learning platform for universities. 12 people, three cities, very different skill sets:

(1) Amsterdam (UTC+1) – **4 people:**

Product managers, UX designers, 1 frontend developer

Strength: user research, product vision, GDPR knowledge

Weakness: no ML or backend expertise

(2) Bangalore (UTC+5:30) - **5 people:**

Backend engineers, ML engineers, 1 DevOps engineer

Strength: recommendation systems, scalable infrastructure

Weakness: no direct contact with end users or customers

(3) Austin (UTC-6) - **3 people:**

Business development, data analyst, customer success

Strength: weekly calls with the pilot customer (University of Texas)

Weakness: no technical or product background

Timezone reality:

Amsterdam <-> Bangalore: 4.5h gap - small morning overlap only

Amsterdam <-> Austin: 7h gap - small evening overlap only

Bangalore <-> Austin: 11.5h gap - nearly zero overlap

The first pilot customer wants a working MVP (Minimum Viable Product) in 9 months.

After the first stakeholder workshops, three problems surface immediately:

(1) Nobody at the university agrees on what the platform should do.

IT wants LMS integration. Professors want content control.

Students want to see why a recommendation was made.

Legal wants FERPA (Family Educational Rights and Privacy Act)

compliance before any student data is processed.

(2) Austin collects new requirements every week in customer calls,

but there is no process to pass them to Amsterdam or Bangalore.

(3) Bangalore has already started building the recommendation engine -

but Amsterdam has not yet decided what data the platform will collect.

Two engineers are frustrated because they keep receiving design mockups

that are architecturally impossible to implement.

The CEO is optimistic. The team is not.

TASK 1 - ANALYSE

a. Which Scrum elements create specific problems here? (Bullets points are fine)

Scrum Element	Still works?	Why / Why not?
2 – week sprints	partially	Sprints can work but time differences may cause problems
Definition of Done	no	Every single customer has a different understanding of done
Backlog prioritization	no	Austin is not putting new processes to the backlog
Sprint Review Demo	yes	Having feedbacks from different shares is vital
Velocity tracking	limited	Velocity may be wrong because of changing requirements

TASK 2 - DECIDE

Three situations. For each: pick one option and justify your choice.

"We would do both" is not an answer.

SITUATION A - WHOSE DESIGN WINS?

Amsterdam has designed a user interface that requires real-time personalisation data streamed from the backend.

Bangalore says this is technically impossible with the current architecture - a full redesign would take 3 months.

You have 9 months until the MVP deadline.

You cannot do a 3-month redesign AND build all features in time.

You have three options:

(1) Amsterdam adapts the design to fit the current architecture.

Risk: the user experience becomes significantly worse.

(2) Bangalore does the redesign. Sprint 1-6 produce almost nothing visible, putting the 9-month deadline at serious risk.

(3) Ship the MVP with the weaker UX (option 1), then redesign after the pilot customer has seen it.

- a. Which do you choose? 3
- b. Who has the authority to make this call - and what happens if Amsterdam and Bangalore cannot agree? Product owner. Decision should be asked to CEO

SITUATION B - THE ONLY MEETING SLOT

You have exactly one 90-minute slot per week where all three teams can meet synchronously (15:00-16:30 Amsterdam time).

This week you have three agenda items that are all critical:

- (1) Finalise the data model - Bangalore is blocked without this.
- (2) Review 6 new requirements from the university - Austin needs a decision before the next customer call on Friday.
- (3) Resolve the Amsterdam/Bangalore architecture conflict from Situation A - it has been unresolved for 2 weeks.

You can cover two of the three properly in 90 minutes.

- a. Which item do you drop from the live meeting?2
- b. How do you handle it differently so it does not stall the team?asynchronous process

SITUATION C - THE LEGAL BLOCKER

Halfway through Sprint 3, Austin calls with urgent news:

the University of Texas legal team has blocked all student data collection until a FERPA (Family Educational Rights and Privacy Act) compliance agreement is signed. Signing takes 6-8 weeks minimum.

Bangalore is currently mid-sprint, building the recommendation engine that depends entirely on that student data.

You have three options:

- (1) Stop Bangalore's work immediately. Wait 6-8 weeks.

Cost: 6-8 weeks of lost development time and team frustration.

- (2) Continue building the engine on synthetic (fake) data.

Risk: the engine may need significant rework once real data arrives and behaves differently than expected.

- (3) Redirect Bangalore to a different part of the product

(e.g., the LMS integration) while legal is resolved.

Risk: context switching costs and the engine falls further behind.

- a. Which option do you choose? 3
- b. How do you communicate this decision to the Bangalore team?

The FERPA approval is an external dependency that we cannot accelerate. Rather than pausing development, we will temporarily shift effort toward LMS integration and platform infrastructure. This keeps the project moving while reducing the risk of building a recommendation engine on assumptions that may later prove incorrect.

TASK 3 - REFLECT

In 2-3 sentences:

- a. Which situation revealed the biggest structural weakness in how LearnLoop is organised? b
- b. What would need to change in the company to make that situation not happen in the first place?

LearnLoop needs a formal requirements and decision-management process. Clear ownership, documented decisions, and structured communication channels would prevent important information from getting stuck between Amsterdam, Austin, and Bangalore.

AI ANALYSIS

Paste into ChatGPT or Claude:

"We are software engineering students. We made process decisions for LearnLoop, a startup with teams in Amsterdam, Bangalore, and Austin.

These were our choices: [paste your Task 2 answers].

Challenge our decisions. For each choice, what is the strongest argument for the option we did NOT pick? Be specific - focus on team dynamics and the 9-month deadline pressure."

Then write 2-3 sentences:

- a. Did the AI make a valid counterargument you had not considered? Or did it ignore a constraint that made your option the only realistic one?
- b. Would you change any of your three choices?